

Sai Jagadeesh Muralikrishnan

Robotics Software Engineer · Perception & Autonomous Systems

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EXPERIENCE

Miso Robotics — Robotics Software Engineer

Sep. 2025 – Present

Pasadena, CA

- Diagnosed and resolved perception-layer defects causing robot-environment collisions and unplanned production downtime, improving deployment reliability across customer kitchen sites.
- Eliminated motion planning failures in constrained environments via site-specific configuration refinement, achieving **100% state-transition success** across all validated scenarios.
- Shipped **8+ ROS 2 behavior primitives** for Flippy's 6-DOF arm covering full cooking-task sequences under real-time constraints; designed HIL test suites catching pre-deployment failures before site rollout.

Kick Robotics — Robotics Engineering Intern

May 2024 – Aug. 2024

Bethesda, MD

- Improved outdoor depth-map precision by **12%** via field calibration of Basler ToF and Intel RealSense D435 cameras, enhancing obstacle-detection reliability for autonomous navigation.
- Reduced end-to-end system latency by **15%** refactoring ROS 2 (C++) perception and control nodes on NVIDIA Jetson; cut QA cycle time **20%** via automated CI/CD with Docker, PyTest, and GoogleTest.

TuTr Hyperloop — Embedded Systems Engineer

Oct. 2022 – Jul. 2023

Chennai, India

- Built multi-sensor fusion pipelines in C++/Python integrated via MATLAB, reducing real-time state-estimation latency by **25%**; achieved **99.9% system uptime** with CAN-based safety-critical VCU signaling.
- Increased HIL and unit test coverage by **40%** through automated Git-based validation pipelines.

PROJECTS

Multi-Agent Robotic Exploration ROS 2, C++, Gazebo, OpenCV, Open3D, PyQt5

Nov. 2023

- Designed and validated MCTS autonomous path planning across **200+ Gazebo simulations**, achieving collision-free multi-robot coverage in unknown environments.
- Built real-time 3D mapping pipeline using OpenCV and Open3D with a live PyQt5 multi-agent operator dashboard.

Text-to-Command Robot Navigation ROS 2, PyTorch, Transformers, LoRA, Gazebo

Nov. 2024

- Fine-tuned T5-Small with LoRA adapters on 24.5K instructions, achieving **98.5% sequence accuracy** for NLP-driven mobile robot navigation; deployed and benchmarked in ROS 2/Gazebo.

CareBotix — AI Patient Monitoring YOLOv8, OpenCV, PyQt6, Flask, MongoDB

Apr. 2025

- **Best Health Track Award**, Morgan Hacks 2025; built real-time YOLOv8-pose detection system with full-stack PyQt6/Flask/MongoDB architecture.

TECHNICAL SKILLS

Perception & Sensors	OpenCV, YOLOv8, PCL, Open3D, RealSense D435, Basler ToF, CUDA, NumPy
Robotics	ROS 2, MoveIt, Nav2, Gazebo, MCTS Planning, CAN Protocols, Embedded C++
ML / AI	PyTorch, TensorFlow, Transformers (HuggingFace), LoRA Fine-tuning, ONNX
Languages	Python, C++, MATLAB, Bash, Shell Scripting, Linux
Infrastructure	Docker, Git, GitHub Actions, CI/CD, AWS, GCP, REST APIs, PyTest, GoogleTest

EDUCATION

University of Maryland — M.Eng. in Robotics — GPA 3.8/4.0

Aug. 2023 – May 2025

College Park, MD

Rajalakshmi Engineering College — B.E. in Mechatronics — GPA 8.7/10.0

Aug. 2018 – Jul. 2022

Chennai, India

NOTABLE ACHIEVEMENTS

Hackathon: 3rd Place & Best Health Track Award — CareBotix, Morgan Hacks 2025

Scholarships: Pathways to PhD & Pathways to Profession — Maryland Robotics Center (2024)

Publication: “Wireless Animatronic Hand Using Infrared Sensor” — ICDSMLA 2021, Springer Nature

Award: Best Paper Award (3rd Place) — Int'l Conference on Data Science, ML & Applications 2021

Leadership: President, COSMO Mechatronics Dept. Club, REC (Apr. 2021 – Apr. 2022)